Department of Electronic & Telecommunication Engineering
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EN4720: Security in Cyber-Physical Systems
Course Project

Milestone 4: Cyber-Attack Detection in a Smart Home System

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1 Introduction

In this milestone, we implement several rule-based anomaly detection functionalities for a smart home system. This can be used to detect unusual activities in the smart home system that could be due to a cyber attack.

This anomaly detection system can be integrated into an existing smart home codebase by calling the `AnomalyDetector.instrument()` function in suitable hot-spots of the program. The instrumentation function has the format shown in Figure 1.

```
def instrument(self,
               eventName: str,      # Event that occured
               userRole: str,       # Role of user that triggered the event (USER/ADMIN/MANAGER)
               userId: str,         # ID of user that triggered the event
               sourceId: str,       # Identifier or IP of device that triggered the event
               timestamp: float,    # Timestamp of event
               context: dict):      # Other parameters with event information (eg. values)
```

Figure 1: Format of the hot-spot instrumentation function

Possible hot-spots to call the instrumentation function are at registration and login endpoints, various sensor and controller APIs, and critical user actions such as rebooting and firmware updates. The detected anomalies will cause an "ALERT" line in the log file with the timestamp and a brief description of the anomaly.

The data given to the instrumentation function is analysed in the following 5 anomaly detection functions:

- Unusual rate checks
- Abnormal value checks for sensor and power reports
- Role-based access anomalies
- Unusual login sources for users
- Sudden variations of device activity

The system has been tested using unit tests for each specific rule, covering both legitimate events and possible cyber attack signatures.

Source code for the anomaly detector is available at: https://github.com/cyber-decryptors/milestone-4

2 Anomaly Detection Functions

This section describes the anomaly detection functions that are implemented and the test cases used to validate the accuracy of flagging.

2.1 Rate Checks

The **checkRateAnomaly** function identifies abnormal behavior by monitoring the frequency of specific events triggered by users or devices within a given timeframe. It is based on the principle that certain cyber-attacks or system faults exhibit patterns of excessive or repetitive activity in a short period.

By detecting events that occur more frequently than a predefined threshold (adjusted based on user roles), the system can raise alerts and prevent further damage. This anomaly detection mechanism is especially effective in spotting brute-force login attempts, account abuse, or rapid toggling of devices.

2.1.1 Rules

Rule 1: Failed Login Attempt Flooding (Role-Based)
Multiple failed login attempts in a short duration may indicate brute-force attacks. The system uses role-based thresholds to minimize false positives and balance security. The conditions used are as follows.

- For ADMIN:
  - Event Type: `login_attempt` with status `failure`
  - Threshold: More than 3 failures within 60 seconds
- For USER:
  - Event Type: `login_attempt` with status `failure`
  - Threshold: More than 5 failures within 60 seconds

Rule 2: Toggle Device Spamming (Role-Based)
Excessive toggling of smart home devices may reflect device misuse, malfunction, or automation abuse. The thresholds differ for users with varying levels of control.

- **For ADMIN:**
  - Event Type: `toggle_device`
  - Threshold: More than 20 toggles within 30 seconds
- **For USER:**
  - Event Type: `toggle_device`
  - Threshold: More than 10 toggles within 30 seconds

Rule 3: Password Change Abuse
Frequent password changes in a short time may indicate suspicious activity. This rule is applied uniformly across roles.

- Event Type: `password_change`
- Threshold: More than 2 changes within 30 minutes

Rule 4: Excessive Device Registration Attempts
Unusual frequency in registering new devices could indicate a security breach or a misconfigured script.

- Event Type: `device_registration`
- Threshold: More than 5 registrations within 1 hour

2.1.2 Test Cases

Case 1: Excessive Device Registration by a User
This test simulates a scenario where the same user attempts to register devices repeatedly. When a user tries to register more than 5 devices within an hour, the system raises an alert. In this case, 6 registration events within 3600 seconds triggered an alert.

```
>>
Testing device registration rate limit
[2025-05-31 22:46:28.719429] ALERT: Too many device_registration events by test_user from src_5 (6 in 36
00 seconds)
.................................................................
```

Figure 2: Rate Check - Device registration flooding

Case 2: Repeated Failed Login Attempts by ADMIN
This test case checks for brute-force attacks targeting an administrator. The system allows up to 3 failed login attempts within 60 seconds. Here, the ADMIN account made 4 failed attempts, resulting in a triggered alert.

```
.----------------------------------------------------------
Testing failed login rate for ADMIN (flag after 4 failures in 60s)
[2025-06-01 15:50:49.224513] ALERT: Too many login_attempt_failure events by admin_fail from src_fail_admin (4 in 60 seconds)
.----------------------------------------------------------
```

Figure 3: Rate Check - Admin failed login attempts

Case 3: Repeated Failed Login Attempts by USER
A regular USER account failed to log in 6 times within a 60-second window. The system threshold for USER failed attempts is 5 per minute. Since it exceeded the limit, an alert was raised.

```
.................................................................
.Testing failed login rate for USER (flag after 6 failures in 60s)
[2025-05-31 21:56:33.719429] ALERT: Too many login_attempt_failure events by user_fail from src_fail (6
in 60 seconds)
.................................................................
```

Figure 4: Rate Check - User failed login attempts

Case 4: Excessive Password Change Requests
The system restricts excessive password change operations to prevent account takeover or abuse. In this test, a user attempted 3 password changes in a short time window (30 minutes), breaching the limit and triggering an alert.

```
..................................................................
.[2025-05-31 22:16:28.735039] ALERT: Too many password_change events by user_pw from src_pw (3 in 1800 s
econds)
.................................................................
```

Figure 5: Rate Check - Excessive password change attempts

Case 5: Toggle Device Spam by ADMIN
An ADMIN user toggled a smart device 21 times within 30 seconds. The system defines this as unusual behavior since the threshold is 20 toggles. Therefore, the system raised an alert.

```
.................................................................
.Testing toggle device spam for ADMIN (flag after 21 toggles in 30s)
[2025-05-31 21:56:48.735039] ALERT: Too many toggle_device events by admin2 from src4 (21 in 30 seconds)
.................................................................
```

Figure 6: Rate Check - Admin device toggle flooding

Case 6: Toggle Device Spam by USER
A USER triggered the toggle event 11 times in just 30 seconds, exceeding the user threshold of 10 toggles per 30 seconds. This case verifies that the system applies rate checks to users regardless of privilege level.

```
.................................................................
.Testing toggle device spam for USER (flag after 11 toggles in 30s)
[2025-05-31 21:56:48.735039] ALERT: Too many toggle_device events by user2 from src3 (11 in 30 seconds)
.................................................................
```

Figure 7: Rate Check - User device toggle flooding

2.2 Value Checks

The **checkValueAnomaly** function identifies anomalous values in smart home sensor readings, specifically focusing on power consumption and temperature. These anomalies often serve as early indicators of compromised devices, hardware failures, or cyber-attacks that seek to manipulate system behavior via forged sensor data.

By tracking historical readings per device, the function identifies deviations that are too abrupt or statistically improbable, such as spikes and drops. These rules are tailored to each sensor type.

2.2.1 Rules

Rule 1: Power Reading Anomalies
Power value anomalies are flagged using the following conditions.

- Event Type: `power_reading`
- Any of the following conditions:
  - Value is negative or zero
  - Value exceeds 150% of historical average (spike)
  - Value is below 50% of historical average (drop)

Rule 2: Temperature Reading Anomalies
Temperature value anomalies are flagged using the following conditions.

- Event Type: `temperature_reading`
- Conditions:
  - Value is out-of-range (< -40°C or > 100°C)
  - Value exceeds 120% of historical average (spike)
  - Value is below 80% of historical average (drop)

2.2.2 Test Cases

Case 1: Normal Power Values
This test verifies that standard power consumption patterns do not raise any false alerts. The values used were: `[100, 105, 110, 95, 102]`. These values are close to each other and within a reasonable

range, resulting in a consistent historical average. As expected, the system did not flag any anomalies during this test.

```
.------------------------------------------------------------
Testing normal power readings (no flag expected)
.------------------------------------------------------------
```

Figure 8: Normal power readings - no alert triggered

Case 2: Abrupt Power Spike
A spike is detected when the power reading exceeds 150% of the previous average. In this test, a consistent sequence of five readings at 100W was followed by an input of 300W. Since 300W is greater than 1.5 times the average of the previous values (average = 100W), it triggered the anomaly detection logic as a significant power spike.

```
.
Testing abrupt power spike (should flag)
[2025-05-31 17:32:29.386210] ALERT: Power spike: 300W > 150% of avg 100.00W from device fan1
.--------------------------------------------------------
```

Figure 9: Power spike detection - alert triggered

Case 3: Abrupt Power Drop
Power drops are flagged when the reading falls below 50% of the recent historical average. In this scenario, five consistent readings of 300W were recorded, establishing an average of 300W. A subsequent input of 0W was then received. Since 0W is significantly below the 50% threshold (i.e., below 150W), the system correctly identified this as a sudden power drop and triggered an alert.

```
--------------------------------------------------------
Testing abrupt power drop (should flag)
[2025-05-31 17:32:29.385884] ALERT: Negative or zero power value 0 reported by device fan1
.--------------------------------------------------------
```

Figure 10: Power drop detection - alert triggered

Case 4: Normal Temperature Values
This test confirms that minor fluctuations in temperature do not trigger alerts. The following readings were used: `[22.0, 21.5, 22.2, 21.8, 22.1]`. These values are close to each other and represent a stable environment. As expected, the system did not flag any anomalies during this test.

```
.------------------------------------------------------------
Testing normal temperature readings (no flag expected)
.------------------------------------------------------------
```

Figure 11: Normal temperature readings - no alert triggered

Case 5: Temperature Spike
A temperature spike is detected when the reading exceeds 120% of the recent average. In this case, five consistent readings of 25.0°C were recorded, resulting in an average of 25.0°C. A sudden reading of 35.0°C was then observed. Since 35.0°C is 140% of the average and exceeds the 120% threshold (30.0°C), the system correctly flagged this as a temperature spike.

```
.----------------------------------------------------------------------
Testing temperature spike (should flag)
[2025-05-31 17:32:30.387913] ALERT: Temperature spike: 35.0°C > 120% of avg 25.00°C from device temp1
```

Figure 12: Temperature spike detection - alert triggered

Case 6: Temperature Drop
A temperature drop is flagged when the reading falls below 80% of the historical average. In this test, five consecutive readings of 30.0°C were recorded, setting an average of 30.0°C. A subsequent reading of 20.0°C was observed. Since 20.0°C is only 66.7% of the average and falls below the 80% threshold (24.0°C), the system correctly identified this as a temperature drop and raised an alert.

```
Testing temperature drop (should flag)
[2025-05-31 17:32:30.387407] ALERT: Temperature drop: 20.0°C < 80% of avg 30.00°C from device temp1
.----------------------------------------------------------------------
```

Figure 13: Temperature drop detection - alert triggered

Case 7: Out-of-Range Temperature
An out-of-range temperature is flagged when the reading exceeds the defined physical or operational limits of the sensor. In this test, two extreme readings were recorded: $-50.0\,^{\circ}\mathrm{C}$ and $120.0\,^{\circ}\mathrm{C}$. Both values fall outside the valid temperature range for the sensor ($-40\,^{\circ}\mathrm{C}$ to $100\,^{\circ}\mathrm{C}$), thus violating the acceptable bounds. The system correctly identified both readings as out-of-range anomalies and raised alerts accordingly.

```
.----------------------------------------------------------------------
Testing out-of-range temperature (should flag)
[2025-05-31 17:32:24.387690] ALERT: Out-of-range temperature: -50.0°C from device temp1
[2025-05-31 17:32:25.387690] ALERT: Out-of-range temperature: 120.0°C from device temp1
.----------------------------------------------------------------------
```

Figure 14: Out-of-range temperature - alert triggered

2.3 Role Aware Filtering

The **checkRoleAwareFiltering** function implements logic to enforce role-based access control and anomaly detection for smart home system events. Below are the rules implemented for role-based access:

2.3.1 Rules

1: After-Hours Login Alert
After-hours logins from unfamiliar devices by high-privilege users could indicate suspicious activity. Therefore the system raises an alert if the following conditions are met.

- Event: `login_attempt`
- Role: ADMIN or MANAGER
- Time: Outside business hours (i.e., before 08:00 a.m or after 18:00 p.m)
- Login from a new source

2: Business-Hour Login Allowance
Logins from new devices during working hours are considered lower risk. Allow the login without flagging it—even if it's from a new source.

- Event: `login_attempt`

- Role: ADMIN or MANAGER
- Time: Within business hours (08:00 to 18:00)

3: No Alert for Regular USER Logins
USERs are presumed to have limited permissions, so their logins pose less risk. Unusual login attempts from new devices will be tracked in rate check implementation. Therefore the system does not flag logins from normal users, regardless of time or source.

- Event: `login_attempt`
- Role: USER

4: Firmware Update Restriction
Firmware updates can drastically affect system behavior and must be restricted. The system raise an ALERT if a non-MANAGER attempts this action.

- Event: `update_firmware`
- Role: not MANAGER

5: Critical Event Restriction
These are high-risk critical operations and must be strictly controlled. Because of that the system raise an ALERT if a non-ADMIN user tries to trigger such critical functions.

- Event: `reboot, disable_alarm or disable_device`
- Role: not ADMIN

2.3.2 Test cases

In these test cases, the INFO log messages are only used for testing purposes and will not be included in a deployed system.

Case 1: test_business_hour_login_no_alert_for_admin
In this test case ADMIN logs into the system at 10:00 AM from a new source. This confirms that business-hour logins from new sources by privileged users are allowed.

```
.TestRoleAwareFiltering.test_business_hour_login_no_alert_for_admin
Testing business-hour login from new source by ADMIN (should not flag)
[2025-05-31 10:00:00] INFO: User 123 with role ADMIN logged in during business hours
[2025-05-31 10:00:00] WARNING: User 123 logged in from new source 012301
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 15: role aware filtering - test case 1

Case 2: test_after_hours_login_alert_for_admin
In this test case ADMIN logs into the system at 10:00 PM from a new source. This ensures after-hour access from unknown sources is flagged.

```
.TestRoleAwareFiltering.test_after_hours_login_alert_for_admin
Testing after-hours login from new source by ADMIN (should flag)
[2025-05-31 22:00:00] ALERT: After-hours login by ADMIN 123 from new source 012302
[2025-05-31 22:00:00] WARNING: User 123 logged in from new source 012302
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 16: role aware filtering - test case 2

Case 3: test_firmware_update_by_non_manager
In this test case USER attempts to execute update_firmware. This validates enforcement of firmware update privilege.

```
.TestRoleAwareFiltering.test_firmware_update_by_non_manager
Testing firmware update by USER (should flag)
[2025-05-31 17:26:29] ALERT: User 124 with role USER tried to update firmware
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 17: role aware filtering - test case 3

Case 4: test_firmware_update_by_manager
In this test case MANAGER executes update_firmware. This confirms legitimate firmware access is allowed without alerts.

```
.TestRoleAwareFiltering.test_firmware_update_by_manager
Testing firmware update by MANAGER (should not flag)
[2025-05-31 17:27:40] INFO: User 125 with role MANAGER updated firmware
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 18: role aware filtering - test case 4

Case 5: test_critical_event_by_non_admin
In this test case USER attempt to trigger disable_alarm. This triggers an alert since only ADMINs can perform this action.

```
.TestRoleAwareFiltering.test_critical_event_by_non_admin
Testing critical event by USER (should flag)
[2025-05-31 17:28:31] ALERT: User 126 with role USER tried to trigger critical event disable_alarm
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 19: role aware filtering - test case 5

Case 6: test_critical_event_by_admin
In this test case ADMIN triggers disable_alarm. This confirms correct permissions for critical operations.

```
.TestRoleAwareFiltering.test_critical_event_by_admin
Testing critical event by ADMIN (should not flag)
[2025-05-31 17:29:56] INFO: User 123 with role ADMIN triggered critical event disable_alarm
.
----------------------------------------------------------------------
Ran 1 test in 0.001s
```

Figure 20: role aware filtering - test case 6

2.4 Unexpected Login Sources

The **checkUnexpectedUser** function implements several rules for handling logins and accesses from new source devices or IPs. This aims to detect possible cyber attacks from external devices while still allowing normal operations, such as users logging in from multiple devices.

2.4.1 Rules

1: Multiple Logins from New Sources
Multiple logins from new sources within a short time interval can indicate that an external attacker has gained access to the smart home system. Therefore, the anomaly detector raises an ALERT flag if the following conditions are met:

- Event: `login_attempt`
- 3 or more logins from different new source IDs
- Within 2 days

In addition, it will show a WARNING log message each time anyone logs in from a new source, for additional safety.

2: Critical Event Triggered after a New Source Login
If a critical event such as a firmware update or reboot is triggered by a user who recently logged in from a new source, the detector will raise an alert since it could be an external attacker trying to disrupt operations. This will minimally affect normal operations since these critical events are rare and not necessary to trigger after logging in from a new device. The conditions for the flag are as follows:

- Events: `reboot, update_firmware, disable_device, disable_alarm`
- Event is triggered within 2 hours of that user logging in from a new source

2.4.2 Test cases

Case 1: test_normal_login_attempts
In this test case, a user logs into the system 3 times from the same source. This confirms that normal, repeated logins by privileged users from the same source do not raise a flag.

```
test_normal_login_attempts (test_anomalies.TestUnexpectedUser.test_normal_login_attempts) ...
Testing normal user logins
ok
```

Figure 21: Unexpected login sources - test case 1

Case 2: test_infrequent_new_sources_no_flag
In this test, a user logs in from 3 different sources with a large time gap between the attempts. There is be a WARNING message to indicate the new source login, but the behaviour is not flagged.

```
-------------
Testing infrequent new source logins
[2025-06-01 12:00:42.961528] WARNING: User 124 logged in from new source 009001
[2025-06-01 17:32:22.961528] WARNING: User 124 logged in from new source 009002
[2025-06-03 19:32:22.961528] WARNING: User 124 logged in from new source 009003
ok
```

Figure 22: Unexpected login sources - test case 2

Case 3: test_multiple_new_sources_trigger_flag
Here, a user logs in from three different sources within a 2-day time window. The third login triggers an ALERT flag, showing that this is suspicious behaviour.

```
-------------------
Testing frequent new source logins
[2025-06-01 11:51:20.084721] WARNING: User 125 logged in from new source 009101
[2025-06-01 11:54:30.084721] WARNING: User 125 logged in from new source 009102
[2025-06-01 11:57:50.084721] WARNING: User 125 logged in from new source 009103
[2025-06-01 11:57:50.084721] ALERT: User 125 logged in from 3 new sources within the last 2 days
ok
```

Figure 23: Unexpected login sources - test case 3

Case 4: test_critical_event_flagged_due_to_recent_new_login
This test checks if a critical action (disabling an alarm) is flagged when it occurs within 2 hours after a login from a new source. The admin's login is followed by a critical event within 30 minutes, and the detector raises an ALERT.

```
--------------------------------------------
Testing critical events triggered after new source login
[2025-06-01 11:51:20.084721] WARNING: User 123 logged in from new source 010301
[2025-06-01 12:21:10.084721] ALERT: Event disable_alarm triggered by user 123 logged in from new source 010301
ok
```

Figure 24: Unexpected login sources - test case 4

Case 5: test_critical_event_not_flagged_if_login_not_recent
This test confirms that critical events are not flagged if they occur a long time after a login from a new source. The ADMIN disables the alarm 3 hours after logging in, and no anomaly is detected.

```
--------------------------------------------
Testing critical events triggered a longer time after new source login
[2025-06-01 11:51:30.084721] WARNING: User 123 logged in from new source 010401
ok
```

Figure 25: Unexpected login sources - test case 5

2.5 Variations in Device Activity Frequency

The **checkActivityIntervals** function implements rules for checking anomalies in the activity frequency of devices and sensors. This includes sudden increases and decreases in frequency as well as inactive periods. The detector considers periodic events such as `device_heartbeat, sensor_report, power_report, data_sync` and establishes an average interval for each unique event and device combination within the first 5 events. It is assumed that the device ID is mentioned in the context variable (source ID is not used for this since the IP addresses of the same sensor might change after a network modification).

2.5.1 Rules

1: Sudden Increase in Event Frequency
A sudden increase in an event frequency from a device can indicate an attack to overload the network or an attempt to cause malfunctions the device. The following conditions detect this situation:

- Intervals between a certain type of event from the same device become 1/10th or less of the average interval
- 5 or more occurrences of the above condition

2: Sudden Decrease in Event Frequency
A sudden decrease in an event frequency from a device can indicate an attempt to cause malfunctions in the device or a sudden increase in the processing load for that device. The following conditions detect this situation:

- Intervals between a certain type of event from the same device become at least 5× larger than the average interval
- 5 or more occurrences of the above condition

3: Inactivity of an Event
If a normally periodic event suddenly stops for a long time, it can indicate that the device was unknowingly disabled by an attacker. Inactivity is detected by the following conditions:

- Checked for all known devices in 1-minute intervals during any call to `instrument()`
- The time passed since the last occurrence of an event from a device is at least 5× larger than the average interval

After inactivity is detected, the next inactivity check for that event is done after a 5× average interval time to avoid flooding the log with a large number of alerts.

4: Update of Event Frequency
In some cases, an update or reset of a device can legitimately change the event frequency. Assuming that this is indicated by an 'update' or 'reset' status in the context variable, the tracked average interval for that event is reset and recalculated.

2.5.2 Test cases

Case 1: test_stable_normal_activity_all_devices
This test ensures that regular heartbeat events from multiple devices at consistent 1-minute intervals do not trigger a flag.

```
test_stable_normal_activity_all_devices (test_anomalies.TestDeviceActivity.test_stable_normal_activity_all_devices) ...
--------------------
Testing stable normal activity (no alerts) for multiple devices
ok
```

Figure 26: Device activity variations - test case 1

Case 2: test_minor_jitter_no_flag
In this test, a single device sends periodic heartbeats with small random jitter of ±5 seconds. These irregularities do not trigger a flag since they are not a significant change in the frequency.

```
test_minor_jitter_no_flag (test_anomalies.TestDeviceActivity.test_minor_jitter_no_flag) ...
Testing small jitter (±5s) in intervals
ok
```

Figure 27: Device activity variations - test case 2

Case 3: test_frequent_activity_flag_after_5
This test simulates a device initially reporting data at a stable 60-second interval. It then starts reporting every 5 seconds. The test has 10 occurrences of rapid reporting, and alerts are raised after the first 5 events.

```
-------------
Testing fast reporting: should be flagged after 5 events
[2025-06-01 12:47:52.139776] ALERT: Sudden increase of reporting rate for sensor_report_controller-4: 60.0 s to 5.0 s
[2025-06-01 12:47:57.139776] ALERT: Sudden increase of reporting rate for sensor_report_controller-4: 60.0 s to 5.0 s
[2025-06-01 12:48:02.139776] ALERT: Sudden increase of reporting rate for sensor_report_controller-4: 60.0 s to 5.0 s
[2025-06-01 12:48:07.139776] ALERT: Sudden increase of reporting rate for sensor_report_controller-4: 60.0 s to 5.0 s
[2025-06-01 12:48:12.139776] ALERT: Sudden increase of reporting rate for sensor_report_controller-4: 60.0 s to 5.0 s
ok
```

Figure 28: Device activity variations - test case 3

Case 4: test_slow_activity_flag_after_5
A device starts with regular power reporting every 60 seconds, and then slows down to report every 400 seconds. Initially this triggers the inactivity flag. After 5 slow events, it also flags the slower reporting rate.

```
-----
Testing slow reporting flagged after 5 events
[2025-06-01 12:54:07.158291] ALERT: Inactivity detected for power_report_motion-sensor-8: typically reports every 60.0 s, inactive for 400.0 s
[2025-06-01 13:00:47.158291] ALERT: Inactivity detected for power_report_motion-sensor-8: typically reports every 60.0 s, inactive for 400.0 s
[2025-06-01 13:07:27.158291] ALERT: Inactivity detected for power_report_motion-sensor-8: typically reports every 60.0 s, inactive for 400.0 s
[2025-06-01 13:14:07.158291] ALERT: Inactivity detected for power_report_motion-sensor-8: typically reports every 60.0 s, inactive for 400.0 s
[2025-06-01 13:20:47.158291] ALERT: Inactivity detected for power_report_motion-sensor-8: typically reports every 60.0 s, inactive for 400.0 s
[2025-06-01 13:20:47.158291] ALERT: Decrease of reporting rate for power_report_motion-sensor-8: 60.0 s to 400.0 s
ok
```

Figure 29: Device activity variations - test case 4

Case 5: test_inactivity_triggers_on_other_device_call
This case verifies that device inactivity is detected during other events from unrelated devices.

```
--------------------------------
Testing inactivity triggered by another device activity
[2025-06-01 12:52:28.149752] ALERT: Inactivity detected for device_heartbeat_temp-sensor-1: typically reports every 60.0 s, inactive for 361.0 s
ok
```

Figure 30: Device activity variations - test case 5

Case 6: test_reset_clears_activity_alert
This test confirms that events with a reset status will recalculate the average activity interval, allowing a legitimate change in the reporting rate. Initially, a decrease in the reporting interval from 60 s to 5 s triggers an alert. Then, an event with reset status occurs, and afterwards, the 5 s reporting interval is not flagged.

```
-------
Testing reset cancels rate deviation alerts
[2025-06-01 12:47:52.156481] ALERT: Sudden increase of reporting rate for data_sync_motion-sensor-8: 60.0 s to 5.0 s
ok
```

Figure 31: Device activity variations - test case 6